SUPERSTORE DATA ANALYSIS

SQL, Power BI, Database Integration

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Dataset Design

Normalization and Structure of Superstore Data Tables

Normalized Tables

The dataset consists of four normalized tables: **Customers**, **Products**, **Geography**, and **Orders**, ensuring efficient data management and retrieval.

Star Schema

A **star schema** design positions the **Orders** table as the central fact table, linking it to dimension tables for enhanced analysis.

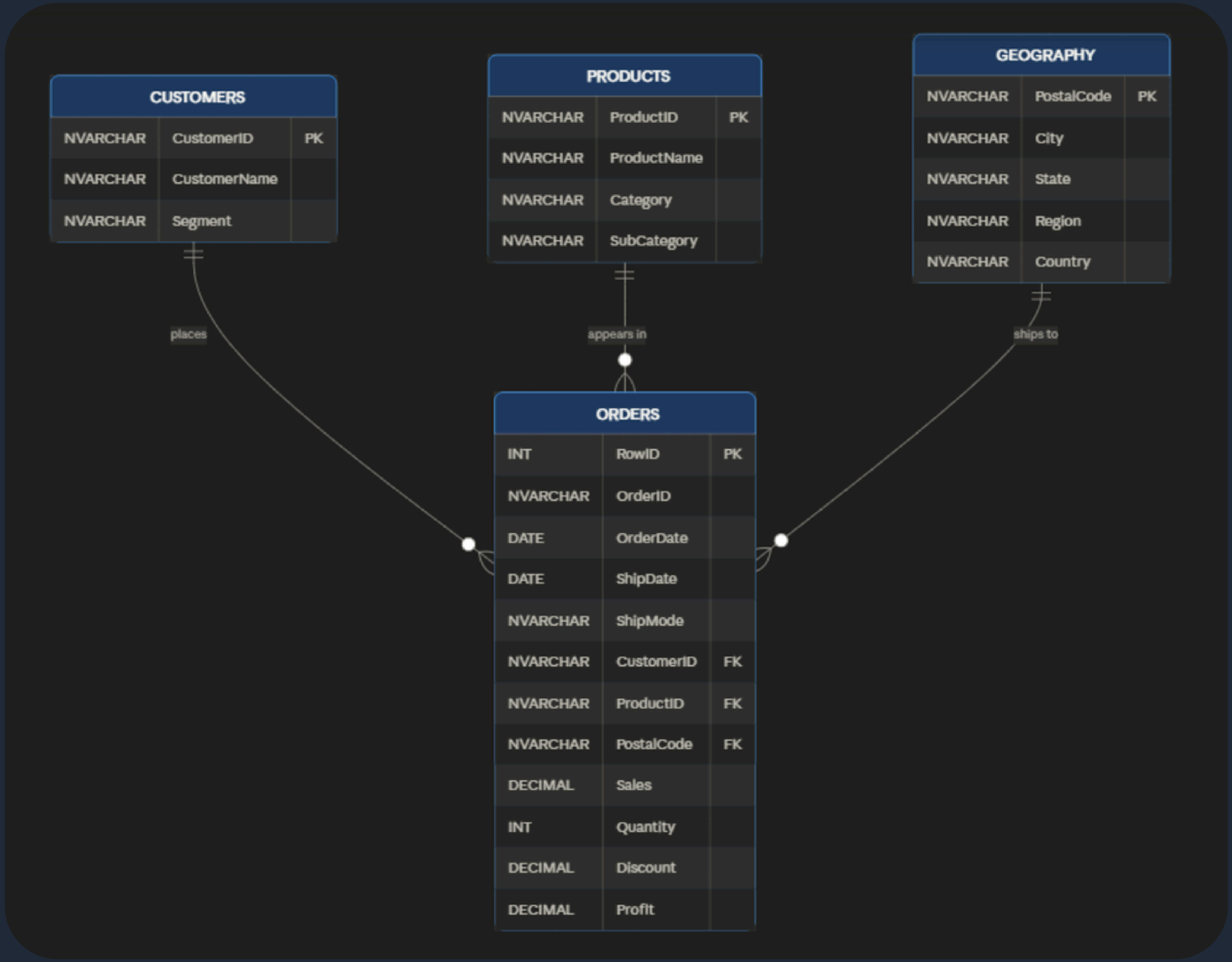
Profit Conversion

The **Profit** field was converted from FLOAT to DECIMAL to prevent rounding errors, ensuring accuracy in financial calculations during analysis.

Entity Relationship Diagram

Linkage of Key Tables

This diagram illustrates the relationships between Customers, Products, and Geography linked to Orders. A Calendar dimension was incorporated in Power BI to enhance time-based analysis capabilities.



SQL Queries Overview

01 Top 10 products ranked by total sales

```
SELECT TOP 10 p.ProductName, SUM(o.Sales) as TotalSales
FROM Orders o
JOIN Products p ON o.ProductID = p.ProductID
GROUP BY p.ProductName
ORDER BY TotalSales DESC;
```

02 Orders shipped via First Class

```
SELECT OrderID, OrderDate, CustomerID
   FROM Orders
   WHERE ShipMode = 'First Class';
```

03 Total profit calculated per state in desc order

```
SELECT g.State, SUM(o.Profit) as TotalProfit
FROM Orders o
JOIN Geography g ON o.PostalCode = g.PostalCode
GROUP BY g.State
ORDER BY TotalProfit DESC;
```

04 Identified loss-making products

```
SELECT p.ProductName, SUM(o.Profit) as TotalLoss
   FROM Orders o
   JOIN Products p ON o.ProductID = p.ProductID
   GROUP BY p.ProductName
   HAVING SUM(o.Profit) < 0;
```

05 Total Quantity Sold by Category

```
SELECT p.Category, SUM(o.Quantity) as TotalQuantity
   FROM Orders o
   JOIN Products p ON o.ProductID = p.ProductID
   GROUP BY p.Category;
```

Power BI Dashboard

Interactive Visuals Overview

The dashboard features interactive visuals including KPI cards, various charts, and a year slicer, allowing dynamic insights into sales data from 2014.

DAX Measures Overview

Total Sales

Total Sales measure calculates the overall revenue from all orders, providing insights into the business's financial performance.

```
Total Sales = SUM(Orders[Sales])
```

Total Profit

Total Profit measure determines the net income from sales, essential for assessing the profitability of the Superstore operations.

```
Total Profit = SUM(Orders[Profit])
```

Profit Margin

Profit Margin is calculated as a percentage, illustrating the ratio of profit to total sales, indicating financial health and efficiency.

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

Sales Last Year

Sales Last Year measure enables comparisons with previous years, essential for understanding growth trends and seasonal fluctuations.

```
Total Profit = SUM(Orders[Profit])
```

Average Order Value

Average Order Value (AOV) gauges the average revenue per transaction, helping to evaluate customer spending and sales strategies.

```
(AOV) Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Orders[OrderID]), 0)

Sales LastYear = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Calendar'[Date]))
```

Key Insights

Profitable States

New York and California emerge as the **most profitable** states, significantly contributing to the overall revenue generated in the project.

Technology Profit

The Technology category consistently generates the **highest profit** per unit sold, indicating strong demand and effective pricing strategies.

Furniture Discounts

Furniture items, particularly tables and bookcases, show persistent losses due to **discount abuse**, necessitating a reevaluation of pricing strategies.

Consumer Segment

The Consumer segment drives **54.95 percent** of total sales, demonstrating its importance in the overall sales strategy and customer targeting.

Q4 Peak Season

The fourth quarter, specifically months 10 to 12, represents the **peak profit season** across all years, highlighting key sales opportunities.

Thank You for Your Attention!

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